scientific data

Article in Press

Check for updates

OPEN

DATA DESCRIPTOR

Hydropower Site-level Monthly Availability Data Ensemble for 1950–2100 at Existing and Potential Global Sites

A. F. M. Kamal Chowdhury[1]✉, Guta Wakbulcho Abeshu[2], Mengqi Zhao[2], Thomas B Wild[3], Hassan Niazi[3], Ying Zhang[3], Gi Joo Kim[4], Matthew Binsted[3], Jonathan Lamontagne[5] & Hong-Yi Li[6]

There is a growing demand for time-series data on hydropower availability at the site level, particularly to enable detailed investigation of the evolving role of hydropower in power grids. Such a dataset for both existing and potential hydropower sites, with global coverage, and incorporating future uncertainty, is still not available. Here, we present HydroSMADE (Hydropower Site-level Monthly Availability Data Ensemble) v1.0, a new open dataset that provides monthly hydropower availability for 1,593 existing and 124,333 potential sites worldwide over the period 1950–2100. The dataset is generated by using a global hydrologic model (Xanthos) with explicit representation of hydropower operations. Specifically, HydroSMADE distinguishes between storage and diversion sites, applies optimized operating rules, and incorporates site-specific characteristics such as generation capacity, maximum turbine flow, and reservoir storage. Driven by bias-corrected meteorological inputs, the site-level hydropower availability data is provided for 30 alternative future scenarios. Technical validation demonstrates consistency with reference datasets and literature across spatial scales, while highlighting improvements in representing operational differences between storage and diversion sites.

Background & Summary

Hydropower plays a crucial role in modern energy systems, serving as both a major source of electricity and a key flexibility resource that balances the intermittency of non-dispatchable generation technologies, such as solar and wind[1]. Hence, its influence on grid stability and reliability is intensifying[2]. Despite hydropower's critical importance, few datasets enable systematic assessment of its operational role, long-term evolution, and reliability under future change globally[3,4]. In particular, power-system studies—often based on production-cost[5–7] and capacity-expansion[1,8,9] models—generally require detailed inputs on hydropower availability (i.e., technically available hydropower that can be dispatched) to capture sub-annual variability (e.g., seasonal) and site-level hydro-physical heterogeneity. Although the emergence of hydropower plant-attribute and reservoir characteristics databases[10–15] and modeling capabilities[4] have spurred several simulation-based datasets, these remain geographically restricted, limited to historical periods, or focused only on existing facilities. For instance, existing datasets provide historical monthly hydropower availability at 0.5° spatial resolution, covering over 90% of globally installed hydropower capacities[14], historical monthly availability for existing sites in the United States[16,17] and Canada[18], and either historical and future projections for existing sites or identification of potential sites in specific countries or regions[3,19–27].

A key gap thus persists: the absence of open, sub-annual hydropower data covering both historical and future periods—and extending beyond existing sites to include potential sites that are often treated as upper bounds

[1]Earth System Science Interdisciplinary Center (ESSIC), University of Maryland, College Park, MD, USA. [2]Pacific Northwest National Laboratory (PNNL), Richland, WA, USA. [3]Joint Global Change Research Institute (JGCRI), Pacific Northwest National Laboratory (PNNL), College Park, MD, USA. [4]ByWater Institute, Tulane University, New Orleans, LA, USA. [5]School of Civil and Environmental Engineering, Tufts University, Medford, MA, USA. [6]Department of Civil and Environmental Engineering and Center for Catastrophe Modeling and Resilience, Lehigh University, Bethlehem, PA, USA. ✉e-mail: kchy@umd.edu

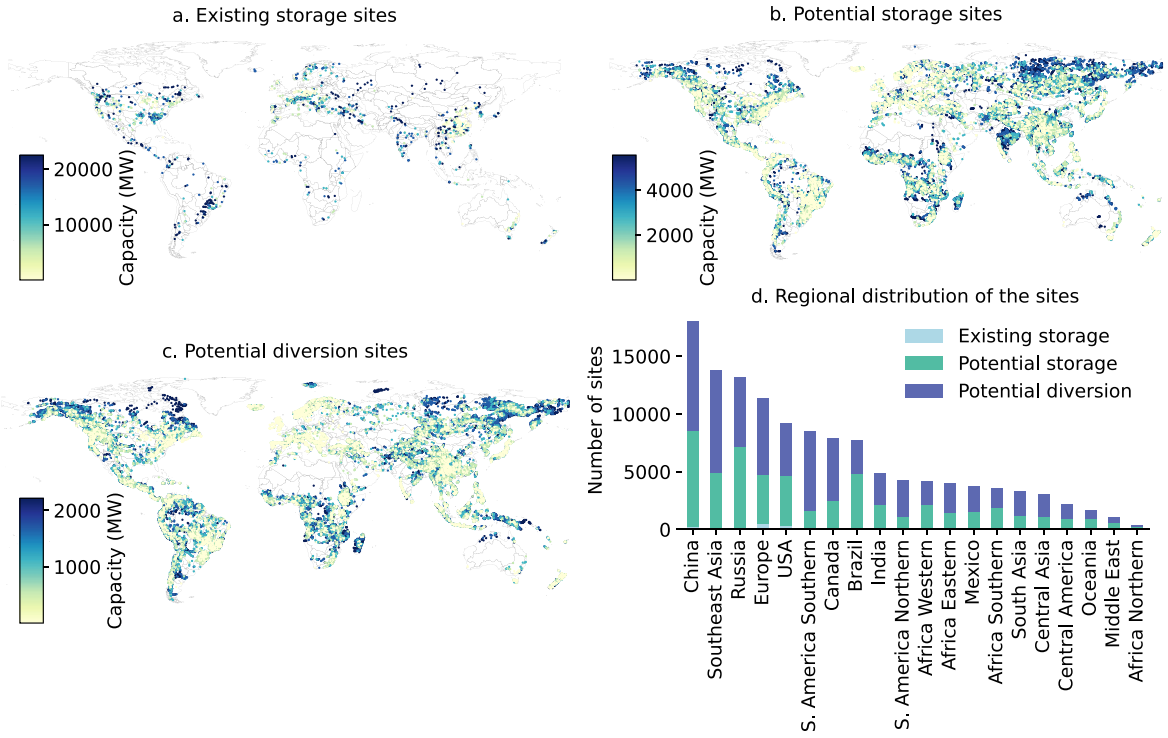


Fig. 1 Locations and hydropower generation capacity of the (**a**) existing and (**b,c**) potential hydropower sites for which monthly hydropower availability is presented in this study. All existing sites are of storage type, while the potential sites include both storage and diversion types. Panel (**d**) shows the regional distribution of the number of sites for each category. Countries included in the multi-country regions are provided in Table S1. The corresponding hydropower capacity in the regions is shown in Figure S1.

of future development in capacity-expansion studies. To fill this gap, we introduce HydroSMADE (v1.0)—a global dataset of monthly hydropower availability spanning 1950–2100, covering an array of existing and technically feasible potential sites (Fig. 1), with future projections generated under 30 alternative scenarios. Beyond power-system modeling at sub-national to global scales, HydroSMADE can support a wide range of applications, including global multi-sector dynamic models that use site-level hydropower availability to construct resource supply curves and assess the long-term evolution of hydropower within a broader human-earth system[28,29].

To develop HydroSMADE, we combined the enhanced capabilities of a state-of-the-art global hydrologic model, Xanthos[30–33], with the recently released global inventory of technically feasible potential hydropower sites by Xu *et al.*[34] (referred to as 'Xu *et al.*' hereafter). Xanthos is an open-source, distributed global hydrologic model that simulates runoff and streamflow at monthly and 0.5° spatial resolution by using meteorological inputs such as precipitation, temperature, wind speed, and solar radiation. Its latest version[30] incorporates explicit representation of hydropower operations (Fig. 2)—a feature that distinguishes it from most global hydrologic models. This capability enables more realistic simulations of streamflow, reservoir operations, and thus hydropower availability. Key inputs include site-level attributes such as plant type (reservoir storage or diversion/run-of-river), location, generation capacity, upstream drainage area, reservoir surface area and volume, and hydraulic head. The 1,593 existing sites with reservoir storage (Fig. 1a) were previously incorporated in Xanthos[30] using plant attributes from Turner *et al.*[35], who compiled the data based on the Global Reservoir and Dam Database (GRanD)[15]. For HydroSMADE, we further incorporated 124,333 potential sites, leveraging the plant attribute dataset of Xu *et al.* About 42% of the potential sites are with reservoir storage (Fig. 1b), while the remainder are diversion sites (Fig. 1c) that use elevation drop or streamflow velocity for power generation. The diversion sites are considered similar to run-of-river, following standard practice[34,36,37]. The global total capacity of the existing storage, potential storage, and potential diversion sites are 495 GW (~40% of globally installed hydropower capacity, as of 2025[38]), 1,811 GW, and 1,008 GW, respectively (Table 1). The regional distribution of the sites is shown in Fig. 1d.

Since Xanthos operates at a 0.5° spatial resolution, we first simulate the hydropower availability at that scale and then disaggregate it to individual sites. The hydropower sites are classified into four categories based on their upstream drainage area (local and large sites) and operational characteristics (storage and diversion): local-storage, large-storage, local-diversion, and large-diversion (see Methods for our approach to distinguishing local from large sites). In the simulation, each Xanthos grid cell can represent one hydropower plant at a time (see further explanation in the *Methods*). We therefore remapped all sites of each type onto the model grid by aggregating key plant attributes—such as reservoir storage volume, generation capacity, and maximum turbine flow—within each cell. Then, we used bespoke operating schemes for the aggregated capacities of each type.

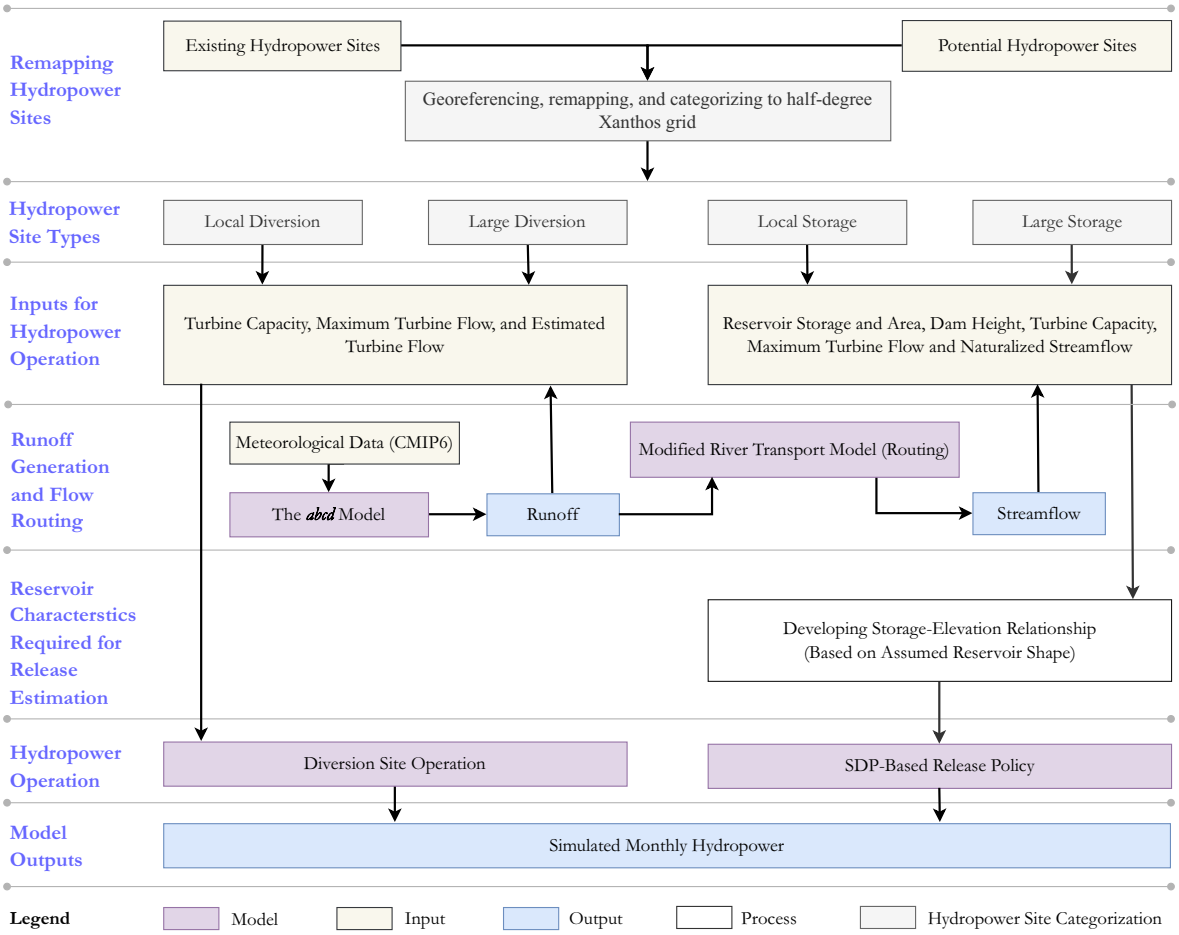


Fig. 2 Schematic of streamflow and hydropower simulation in the global hydrologic model, Xanthos.

Status	Type	Sites	Capacity (GW)
Existing	Large Storage	887	444
Existing	Local Storage	706	51
Potential	Large Storage	28,129	1,611
Potential	Local Storage	24,480	200
Potential	Large Diversion	13,957	383
Potential	Local Diversion	57,767	625
	Total	**1,25,926**	**3,313**

Table 1. Summary of the existing and potential hydropower sites and corresponding capacities included in HydroSMADE v1.0. The sites are divided into four categories based on their upstream drainage area (local and large sites) and operational characteristics (storage and diversion). Note that the included existing sites are only of 'storage' type.

We simulated the operations of storage-type sites using a stochastic dynamic programming (SDP) approach, which optimizes monthly release policies to maximize hydropower generation as a function of storage level (see *Methods*). For diversion (run-of-river) sites, we determined release as a function of monthly inflow, accounting for operational constraints and downstream flow requirements. This produced monthly hydropower availability at the grid scale, which we subsequently disaggregated to the site level according to the ratio of generation capacity for each site aggregated within a cell.

HydroSMADE v1.0 provides an ensemble of 30 alternative scenario-based projections of future hydropower generation through 2100. Each scenario captures a physically plausible alternative trajectory of future hydrologic change (i.e., runoff, streamflow). These futures are produced by forcing Xanthos with 30 alternative hydrometeorological projections from the sixth phase of the Coupled Model Intercomparison Project (CMIP6). Specifically, we simulated the full factorial combination of three standard CMIP6 atmospheric forcing pathways (SSP1-2.6, SSP3-7.0, and SSP5-8.5) and ten CMIP6 General Circulation Models (GCMs). The ten GCMs we use here include: GFDL-ESM4, IPSL-CM6A-LR, MPI-ESM1-2-HR, MRI-ESM2-0, EC-Earth3, CanESM5,

MIROC6, CNRM-ESM2-1, UKESM1-0-LL, and CNRM-CM6-1. Xanthos is forced with the following hydrometeorological variables from the GCMs: precipitation, air temperature, minimum air temperature, relative humidity, shortwave radiation, longwave radiation, and wind speed. We use the product of the Inter-Sectoral Impact Model Intercomparison Project phase 3b (ISIMIP3b), which provides bias-adjusted and statistically downscaled (by using a trend-preserving method, presented in ref. [39]) data of the hydrometeorological variables for the three forcing pathways and ten GCMs. Since the hydrometeorological data for other forcing pathways and GCMs are not available from ISIMP3b[40], those scenarios are not included in HydroSMADE. Subsequent sections describe our methods, technical validation, and data records, together with guidance for potential applications and limitations.

Methods

Hydropower simulation using Xanthos. Xanthos is an open-source, distributed global hydrologic model with nearly a decade of continuous development[30–33]. Although originally designed to support the Global Change Analysis Model (GCAM)[41] and used in a wide range of global analyses (e.g., ref. [28,29,42–46]), it has also been used in diverse regional to sub-national studies (e.g., ref. [9,47–51]). Operating at a 0.5° spatial resolution and a monthly time step, Xanthos simulates runoff and streamflow with global coverage, while its latest version[30] also introduces the representation of hydropower operations (Fig. 2). The model has three core components: runoff generation, river routing, and hydropower operations, with the latter providing dedicated schemes for operations of storage and diversion sites. We briefly describe the key components of Xanthos below, while more details on its setup, inputs, and outputs are available at https://github.com/JGCRI/xanthos. The key datasets and model configurations used in the Xanthos simulation for HydroSMADE v1.0 are specified in Supplementary Table S2.

Runoff-generation. Xanthos simulates runoff using the "*abcd*" water balance model, originally developed by ref. [52] and later integrated into Xanthos by ref.[33]. Driven by meteorological inputs such as precipitation and temperature, the model estimates direct runoff, baseflow, evapotranspiration, and soil moisture at a monthly time step for each 0.5° grid cell. The formulation involves five key parameters that represent critical soil–hydrologic and hydrometeorologic characteristics, including the propensity of runoff to occur before the soil is fully saturated, the maximum capacity for evapotranspiration and soil moisture storage, the fraction of water that recharges groundwater, the rate at which groundwater is released as baseflow, and snowmelt dynamics.

River-routing. To represent the flow of water through river networks, Xanthos partitions the global land surface into 235 major basins and routes streamflow between grid cells using a cell-to-cell transport scheme based on the modified river transport model (MRTM)[53,54]. At each grid cell, total runoff—the sum of direct runoff and baseflow—joins the main channel as lateral flow. Flow in the channel is then routed downstream through a linear channel storage routing approach. The runoff and streamflow outputs were subsequently calibrated and validated against historical data to refine parameters and assess model fidelity (see 'Technical Validation', and supplementary Text S1 and Figures S2-S4).

Hydropower operation. To create HydroSMADE, storage and diversion hydropower sites are simulated using distinct operating schemes, which are described in the following two sub-sections.

Operation of storage hydropower sites. For the hydropower sites with reservoir storage, similarly to ref. [30], we utilize an SDP-based hydropower operation scheme[35,55] that makes sequential decisions across temporal stages, with an objective to maximize annual hydropower generation. Monthly release decisions are obtained by solving the following objective function *f(S,i)*:

$$f_{m,y}(S_{m,y}, i_{m,y}) = \max_{R_{m,y}} E\{B_{m,y}(S_{m,y}, i_{m,y}, R_{m,y}) + f_{m+1,\ y}(S_{m+1,\ y}, i_{m,\ y})\} \quad \forall\ S_{m,y}, i_{m,y}1$$

where $R_{m,y}$ is the release in month m of year y, determined as a function of storage $S_{m,y}$ and inflow $i_{m,y}$. Release in the current and successive months is maximized through immediate benefit $B_{m,y}(S_{m,y}, i_{m,y}, R_{m,y})$ and future benefit functions $f_{m+1,y}(S_{m+1,y}, i_{m,y})$, respectively. For each storage–inflow state, $R_{m,y}$ is selected to maximize the sum of the immediate hydropower benefit $B_{m,y}$ and the expected future benefit $f_{m+1,y}$. The immediate benefit represents hydropower production during the current month and is proportional to $R_{m,y}H_{m,y}$, as described by Eq. (2). The future-benefit function represents the maximum expected hydropower production obtainable from the succeeding reservoir state after the current release decision. The expectation accounts for uncertainty in the reservoir inflows used to train the SDP release policy. E is the expectation over the current month. While a minimum environmental flow is applied for release below 10% of the mean annual inflow, consistent with established practice[56,57], the release is set to the maximum turbine flow (i.e., the maximum allowable flow rate through the turbine) when the reservoir storage level reaches its maximum capacity. Once the release R (in discharge units of m^3s^{-1}) is determined, the hydropower availability P (in kilowatts) is estimated by Eq. (2), where H is the turbine head (m), ρg is the specific weight of water (kNm^{-3}), and η is an efficiency factor (set to 0.9).

$$P = \eta \times \rho g \times R \times H \tag{2}$$

The hydropower generation is maximized by maximizing $R \times H$, given that the rest of the terms are constants. The maximization is obtained through the following steps. First, the turbine flow and the storage level are discretized into 10 and 4000 equal increments, starting from zero to the maximum turbine flow and the

reservoir storage capacity, respectively. Second, a depth-volume relationship is established by assuming a wedge shape for each reservoir, which provides storage depth corresponding to each discretized storage level. Then, the turbine head (H) is estimated by adding the storage depth to the intake elevation, where the latter is estimated as the difference between the maximum turbine head and the maximum storage depth. The maximum turbine head (m) is calculated from the hydropower plant's generation capacity (kilowatts) and maximum turbine flow (m^3s^{-1}) as:

$$H_{\max} = (generation\ capacity)/(\rho g \times maximum\ turbine\ flow \times \eta) \quad (3)$$

whereas the maximum storage depth is assumed to be equal to the dam height. These two steps result in a 4000×10 matrix of $R \times H$ for each month, from which the best monthly release policy is determined by backward recursive iterations (e.g., from December to January) to yield maximum hydropower generation over a year. To ensure robust convergence, the implementation incorporates adaptive refinement mechanisms where discretization levels and convergence tolerances are iteratively adjusted if initial policy convergence criteria are not met within the maximum iteration limit.

Operations of diversion hydropower sites. For diversion hydropower sites, a different approach is employed. Unlike storage facilities, diversion hydropower sites operate as run-of-river systems without significant storage capacity for flow regulation. The operation strategy focuses on maintaining environmental flow requirements while maximizing power generation from available river flow. Monthly power generation is determined through a simplified flow allocation process that accounts for ecological constraints and turbine limitations. The operational framework begins with environmental flow protection, where the minimum ecological flow (Q_e) is established as the 30th percentile of historical monthly flows at each site, similar to Xu *et al.* This conservative approach ensures adequate environmental protection across varying hydrological conditions. The available flow for power generation (Q_t) is then calculated as:

$$Q_t = (Q_{river} - Q_e) \times 0.5 \quad (4)$$

where Q_{river} is the natural river flow and the 0.5 factor represents a conservative utilization rate, accounting for operational constraints[34]. Negative values are set to zero to prevent flow reversal. The diversion capacity (Q_d) for each site is determined as the 97th percentile of the available turbine flow (Q_t) across the historical period, representing the practical upper limit of flow diversion under most operational conditions[34]. During operation, the actual flow through the turbines is constrained by the maximum turbine flow, obtained from Xu *et al.* Power generation follows the same fundamental relationship as storage facilities as in Eq. (2), where the constrained turbine flow is used as the release (R). The turbine head (H) is calculated from the generation capacity and maximum turbine flow to ensure consistency with design specifications. This simplified operational framework eliminates the need for complex optimization algorithms while ensuring environmental protection and realistic power generation estimates. The approach accounts for the operational reality that diversion facilities must work within the natural flow regime while maintaining ecological integrity of the river system.

Mapping hydropower sites into Xanthos grids. Since Xanthos runs at 0.5° spatial resolution, we aggregated the existing and potential hydropower sites of each category into the overlaying Xanthos grids, simulated hydropower availability at the grid level, and then disaggregated them to the site level.

To ensure accurate spatial representation and operational modeling, sites were mapped into grids through three main steps: georeferencing, categorization, and aggregation. First, each hydropower site was assigned to the corresponding Xanthos grid cell based on its geographic coordinates. Second, sites were classified using a two-tier system. In the primary tier, sites with reservoir storage capacity were categorized as 'storage', while those without storage capacity were categorized as 'diversion' facilities. In the secondary tier, sites within each category were further distinguished as 'local' or 'large' depending on their upstream drainage area, reported in Xu *et al.*, relative to the drainage area of the assigned grid cell. Sites with upstream drainage area ≤ grid cell drainage area were labeled as *local*, while those with larger upstream drainage areas were labeled as *large*. The spatial remapping process successfully located nearly all existing and technically feasible potential sites within the Xanthos grid framework. Only 428 potential sites (0.34% of the total 124,761 potential sites collected from Xu *et al.*), for which we could not match an overlaying Xanthos grid with equivalent drainage area, were not included in HydroSMADE. Among the mapped sites, 44% of existing storage, 47% of potential storage, and 81% of potential diversion sites are classified as local, with the remainder categorized as large sites (Table 1). For large sites, we further evaluated the consistency between reported upstream drainage areas and corresponding Xanthos grid cell upstream drainage areas. Sites with discrepancies exceeding 20% underwent spatial readjustment, where they were systematically relocated to neighboring grid cells (within 0.5° and 1.0° search radii) that minimized drainage area mismatches while maintaining basin boundary constraints. With respect to the Xanthos model structure, local sites are placed on tributary reaches within each grid cell, whereas large sites are located on the main-stem river.

Third, following georeferencing and categorization, sites were aggregated at the grid cell level to create representative hydropower facilities for model integration. Multiple sites of each category falling within the same Xanthos grid cell were consolidated using variable-specific aggregation rules. The generation capacity was summed across sites to represent the total potential within each grid cell. Performance characteristics such as head and maximum turbine flow were initially aggregated using the maximum site values to construct a representative facility for each site category within each grid cell. For storage facilities, reservoir surface area and storage capacity were summed to represent total storage potential per grid cell. To ensure physical consistency

in the aggregated data, the power Eq. (2) was applied to verify that power, flow, and head relationships remained consistent after aggregation. When discrepancies were identified, head values were adjusted to maintain equation balance while preserving the aggregated flow and power values, as these directly represent the summed potential of individual sites.

The aggregation process resulted in four datasets of aggregated sites at the grid level: 10,948 grid cells with local diversion sites, 6,828 with local storage sites, 8,061 with large diversion sites, and 10,065 with large storage sites. Then, Xanthos' hydropower module was run separately for each of the four site types, resulting in four datasets of monthly hydropower availability at the grid level. Notably, we did not consider the cascading effects of sites within a grid cell, nor the competitive use of water among multiple site types and other water users (e.g., water demands for irrigation or municipal use). This is because Xanthos was designed to simulate the operation of only one reservoir per grid cell at a time, as a cell can receive only one routed inflow. Although in ref. [30], Xanthos was used to simulate the operations of both the hydropower and non-hydropower reservoirs, the independent operational type for a grid cell was assigned based on the aggregated storage capacity of each type of reservoir within the cell (e.g., a grid cell was classified as hydropower if the total storage of hydropower reservoirs exceeded that of non-hydropower reservoirs). Accordingly, in producing HydroSMADE data, the individual runs for each of the four site-types were independent of the operations of other types of hydropower sites and non-hydropower reservoirs. As a result, the hydropower availability in HydroSMADE reflects an optimized estimate of generation potential under given hydrometeorologic conditions, independent of competing water demands.

Finally, we disaggregate the grid-level hydropower availability to individual sites in proportion to their generation capacity within each grid cell. Because sites are grouped by operational type during aggregation (e.g., sites of the 'large storage' category under a grid cell are aggregated into one), we assume that a site's share of the grid-level hydropower availability is directly proportional to its share of the total generation capacity for that type.

Scenario ensemble. To create the 30-member ensemble of HydroSMADE v1.0, Xanthos was forced by the bias-adjusted meteorological forcing for each SSP-RCP-GCM pairing mentioned above. This enabled us to simulate 30 scenarios of runoff and streamflow, and thus the hydropower availability for 1950–2100. The historical data for the 1950–2015 period follow a consistent pattern across scenarios, while the data for the 2016–2100 period represent an ensemble of varying hydropower availability under alternative future climatic and socio-economic conditions. The same previously calibrated basin-specific Xanthos parameterization, routing configuration, and hydropower-site inputs were used for all 30 simulations. The ensemble members differ in their ISIMIP3b meteorological forcing.

Data Records

The HydroSMADE v1.0 dataset[58] is publicly available on MSD LIVE at https://doi.org/10.57931/3015845. It consists of 122 files: a text file (readme.txt) containing a brief description of the included data, a CSV file containing site attributes, and the remaining 120 files (in CSV) containing site-level monthly hydropower availability.

The site attribute file (site_attributes.csv) assigns a unique identifier to each existing and potential hydropower site included in HydroSMADE v1.0. It also provides detailed attributes, including geographic location, status (existing or potential), type of site based on mapping category (local or large) and operational type (storage or diversion), generation capacity (MW), maximum turbine flow, upstream drainage area, dam height, reservoir area and volume (for storage sites only), and capital and O&M costs (for potential sites only). Attribute data for potential sites are from Xu *et al.*, while those for existing sites are drawn from Turner *et al.*[35], who compiled them based on GRanD[15]. To facilitate basin-level analyses, each site is also mapped to one of 235 basins in Xanthos.

The site-level monthly hydropower availability data (in MW) for each of the 30 scenarios are provided in four separate CSV files for the four types of sites: large_storage, local_storage, large_diversion, and local_diversion. The CSV files are named using the format: "site_level_monthly_hydro_mw_1950_2100_<site type>_<SSP_RCP_GCM>.csv". In each file, the first column lists the site identifiers, which can be used to link the data with the site attributes. The subsequent columns provide monthly hydropower availability from 1950 to 2100, with column headers specifying the time stamps of the monthly records.

Technical Validation

To validate HydroSMADE v1.0, we begin by assessing its performance at the annual scale across all global sites, examining both historical representation and projected future trajectories. This approach is necessitated by the lack of reference data at a monthly resolution for the large number of sites included in the dataset. We then conduct a more detailed analysis for eight major river basins spanning diverse geographic regions, focusing on annual trends and monthly variability, and benchmarking our results against relevant literature. Finally, for a set of existing hydropower sites across the United States, we compare HydroSMADE with RectifHydPlus[16], a monthly hydropower reconstruction dataset for 1980-2019.

HydroSMADE v1.0 was produced with Xanthos v2.4.1 using an existing basin-specific parameterization. This parameterization was calibrated against WaterGAP2-2e monthly runoff[57,59] for 1901–1990 and evaluated against the same reference product for 1991–2019. Across the 235 Xanthos basins, median Kling–Gupta efficiency[60,61] was approximately 0.8 for both annual and monthly runoff during the calibration and evaluation periods, although performance varied among basins (Supplementary Text S1 and Figures S2–S4). Overall, these results indicate that the calibrated runoff parameterization reproduced the annual and monthly variability of the WaterGAP2-2e reference with satisfactory aggregate performance across the 235 basins. Because WaterGAP2-2e is itself a model-based runoff product, this comparison evaluates consistency with

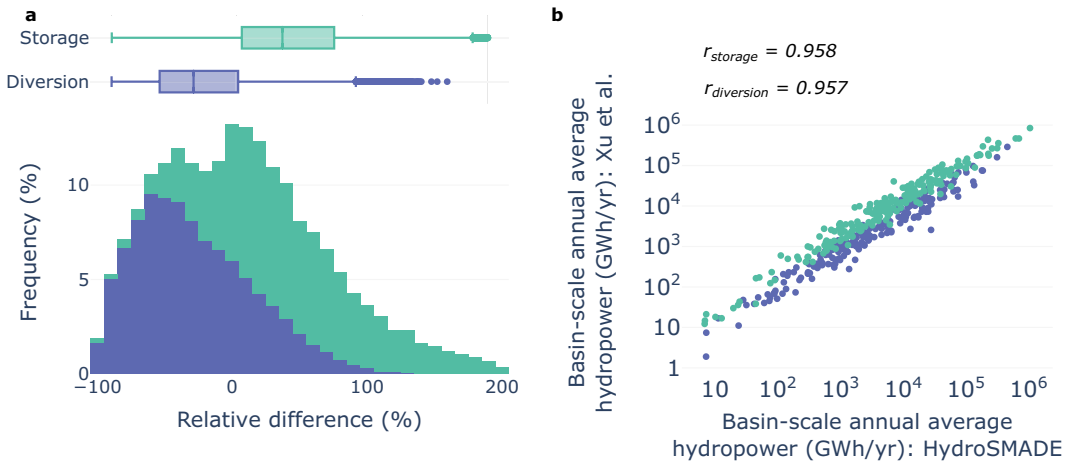


Fig. 3 (**a**) Distribution of relative differences in site-level annual average hydropower availability (1979–2016) between Xu *et al.* and HydroSMADE. For each site, the relative difference is calculated as (HydroSMADE mean − Xu *et al.*)/Xu *et al.*, where the HydroSMADE mean is the ensemble average across 30 scenarios. Positive values indicate that HydroSMADE estimates are higher than Xu *et al.*, and negative values are the opposite. Results for storage and diversion sites are shown in separate colors, with boxplots indicating quartiles and the full range. (**b**) Basin-scale comparison of annual average hydropower availability (1979–2016) between Xu *et al.* and HydroSMADE (ensemble mean of 30 scenarios).

that hydrologic reference rather than agreement with observations. Separately, Abeshu *et al.*[30] evaluated the enhanced Xanthos water-management framework against observed monthly streamflow at 91 large-basin outlets and found improved performance in 75 basins.

Historical representation. We validated HydroSMADE by comparing annual average hydropower availability for 1979–2016 at all global potential sites against the corresponding dataset from Xu *et al.* Relative differences were calculated between the ensemble mean of HydroSMADE's 30 scenarios and the Xu *et al.* data, where positive values indicate HydroSMADE estimates are higher, negative values lower, and zero indicates perfect agreement. Figure 3a shows the distribution of relative differences for potential storage and diversion sites. While the overall median relative difference across all sites is close to zero (Supplementary Fig. S5), medians for storage and diversion sites are approximately +35% and −35%, respectively. This indicates that HydroSMADE tends to estimate higher availability for storage sites and lower availability for diversion sites compared to Xu *et al.* Results are broadly consistent across regions, though the magnitude of differences varies geographically (Supplementary Fig. S6).

These systematic differences likely arise from how hydropower operations are represented in the two datasets. In HydroSMADE, Xanthos simulates storage-site operations endogenously using stochastic dynamic programming, with monthly releases optimized to maximize annual hydropower generation (see *Methods*). Diversion sites are also operated under an energy-maximization rule, but generation depends only on elevation drop or streamflow velocity, which amplifies seasonal variability. By contrast, Xu *et al.* estimate hydropower availability from mean daily discharge, derived from a 0.05° daily discharge records for 2.89 million global rivers over the 1979–2016 period[62], without explicitly modeling storage dynamics or seasonal variability in diversion sites (see Supplementary Text S2). As a result, diversion-site capacity factors in Xu *et al.* are nearly identical to, or even higher than, those of storage sites (Table 2), whereas HydroSMADE produces higher capacity factors for storage sites—a pattern consistent with literature[63]. Although site-level relative differences span a wide range, basin-scale totals show strong agreement between HydroSMADE and Xu *et al.* datasets, with correlation coefficients of ~0.96 for both storage and diversion sites (Fig. 3b).

Projected future trajectories. We next evaluate projected changes in hydropower availability using HydroSMADE. Specifically, we compare basin-scale changes in annual mean hydropower availability between a historical baseline (1970–2000) and a future period (2070–2100) against results from Gernaat *et al.*[64] (referred to as 'Gernaat *et al.*' thereafter) The latter provides relative changes for 158 basins, averaged across four GCMs (GFDL-ESM2M, HadGEM2-ES, IPSL-CM5A-LR, and MIROC5) under RCP2.6 (Fig. 4a). We contrast these with HydroSMADE's multi-model mean of 10 GCMs under SSP1-2.6 for storage (Fig. 4b) and diversion (Fig. 4c) sites.

Across most basins, HydroSMADE and Gernaat *et al.* show consistent directional trends: for example, decreasing hydropower availability in much of South America and Southern Africa, and increasing availability in Northern Africa, South Asia, and Southeast Asia. Overall, HydroSMADE trajectories agree with Gernaat *et al.* for ~70% of the 158 basins (Fig. 5a,b). However, the magnitude of change is often larger in HydroSMADE, with future shifts ranging from −13% to 36% for storage sites and −19% to 155% (over 50% for only eight basins) for diversion sites, compared to −8% to 13% in Gernaat *et al.* As a result, correlations between the two datasets are modest. HydroSMADE's wider range is likely because we included more GCMs than Gernaat *et al.*, while we also modeled storage and diversion type sites with separate operating schemes. Moreover, HydroSMADE's stronger

Basin	Type	Capacity (GW)	Annual average hydropower (TWh/yr): Xu et al.	Annual average hydropower (TWh/yr): this study	Relative difference (%)	Capacity factor: Xu et al.	Capacity factor: this study
Amazon	Diversion	64	314	160	−49	0.56	0.28
	Storage	40	151	178	18	0.44	0.51
Ganges_Bramaputra	Diversion	112	458	289	−37	0.47	0.29
	Storage	157	623	467	−25	0.45	0.34
Indus	Diversion	49	186	75	−60	0.44	0.18
	Storage	85	329	358	9	0.44	0.48
Mekong	Diversion	15	63	39	−39	0.47	0.29
	Storage	85	316	269	−15	0.42	0.36
Nile	Diversion	11	57	31	−45	0.58	0.32
	Storage	41	169	182	7	0.48	0.51
Pacific_Northwest_Basin	Diversion	17	61	36	−41	0.42	0.25
	Storage	51	104	148	42	0.23	0.33
Salween	Diversion	10	43	21	−51	0.49	0.24
	Storage	152	702	466	−34	0.53	0.35
Zambezi	Diversion	6	22	13	−39	0.4	0.25
	Storage	43	163	190	16	0.44	0.51

Table 2. Comparison of basin-scale annual average capacity factors (1979–2016) between HydroSMADE (ensemble mean of 30 scenarios) and Xu *et al*. Results are shown for eight representative basins spanning diverse geographic regions (see basin locations in Figure S7).

signals are closely tied to basin-scale changes in streamflow simulated by Xanthos (Fig. 4d), with hydropower availability strongly correlated with corresponding streamflow changes, ranging from −19% to 82% (Fig. 5c,d).

Annual trend and seasonal variability. Future changes in streamflow and hydropower availability highlight several global hotspots for hydropower shifts (Supplementary Figs. S7-S8). We focus on eight representative basins—the Pacific Northwest, Amazon, Zambezi, Nile, Indus, Ganges-Brahmaputra, Salween, and Mekong—where we examine long-term annual trends (Fig. 6) and seasonal variability (Fig. 7). Annual trajectories are shown for 1950–2100, with GCM-based uncertainty ranges under three hydrometeorologic futures. Seasonal variability is presented for a historical baseline (1970–2000) and a future period (2070–2100), with the latter including uncertainty across 10 GCMs under the three futures.

In the Pacific Northwest Basin, HydroSMADE projects a slight decline in annual hydropower availability at storage sites (Fig. 6a), while diversion sites show a small increase (Fig. 6b). This pattern reflects shifts in seasonality under warming conditions, where reduced winter snowpack and earlier spring melt alter runoff timing—shifting flows from late spring and summer into winter or early spring and reducing late-summer flows[65,66]. Consistent with this, HydroSMADE shows declines in July-November generation at both site types (Fig. 7). Diversion sites benefit from higher December-May flows, whereas storage sites do not, due to summer drawdown that constrains reservoir refill. Despite climate projection uncertainties, this seasonal shift aligns with recent studies[66,67].

For the Amazon and Zambezi, HydroSMADE indicates declining trends at both storage and diversion sites (Fig. 6), with consistently lower production across all seasons (Fig. 7). These results agree with Gernaat *et al*. (Table 3) and with basin-specific studies for the Amazon[68] and Zambezi[69]. The Nile Basin shows an increasing trend in future hydropower availability, consistent with Gernaat *et al*. and with the synthesis by Wasti *et al*.[70]. In South and Southeast Asia, the Indus, Ganges-Brahmaputra, Mekong, and Salween basins all exhibit increasing trends in HydroSMADE, also consistent with Gernaat *et al*. The magnitude of these changes appears largely driven by basin-scale hydrologic shifts (Table 3 and Supplementary Fig. S9). Regional studies[71–73] similarly project rising streamflow in these basins due to intensifying precipitation and glacier melt.

Evaluation of monthly hydropower availability. Because HydroSMADE provides monthly estimates, its seasonal distribution should be assessed directly at the monthly scale rather than inferred from annual- or basin-scale comparisons. The results below support HydroSMADE as a reasonable representation of typical within-year hydropower availability for its intended national, continental, and global applications, while also showing that its monthly profiles should not be interpreted as reliable reconstructions of individual-facility operations. This distinction reflects the dataset's purpose. HydroSMADE estimates technically achievable availability under modeled, energy-maximizing operations for existing and prospective sites, including under future hydrologic conditions. Many prospective sites have no generation records or operating rules, and future operations at existing facilities may differ from historical practice. Reproducing historical dispatch is therefore neither possible across the dataset nor the appropriate general validation target. Following ref. [74], we view this validation as a "fit-for-purpose" assessment: whether HydroSMADE adequately represents the characteristics of the system that matter for its intended application. The underlying reservoir-routing and hydropower-management framework has been evaluated previously in Abeshu *et al*.[30]; here we ask the narrower, complementary question of whether the final product provides a reasonable typical distribution across months.

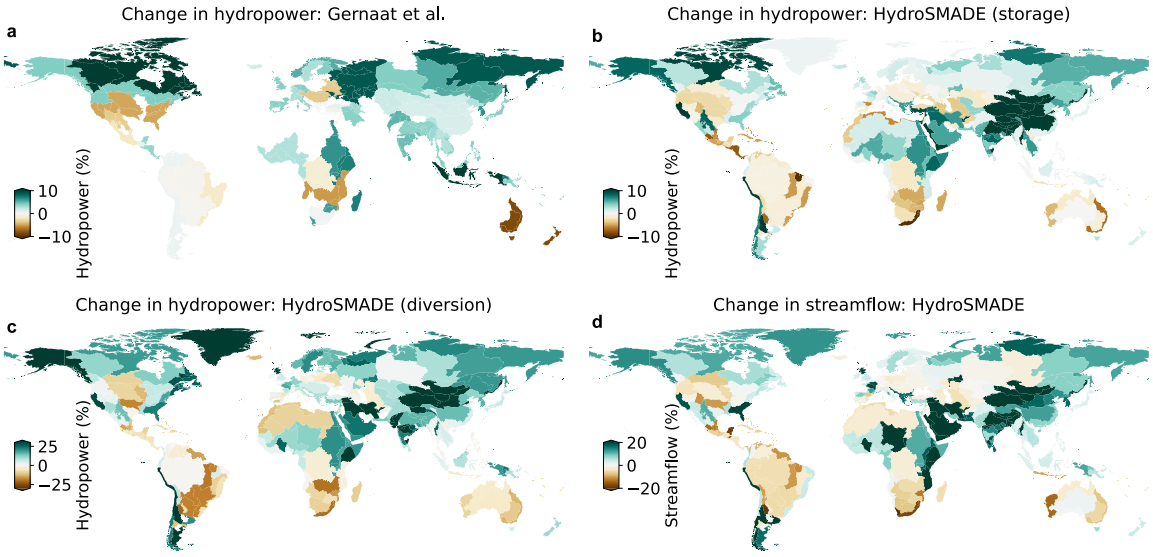


Fig. 4 Basin-scale changes in annual average hydropower availability and streamflow between a historical baseline (1970–2000) and a future period (2070–2100). Changes in: (**a**) annual average hydropower availability under RCP2.6 from Gernaat *et al.*, averaged across four GCMs (GFDL-ESM2M, HadGEM2-ES, IPSL-CM5A-LR, and MIROC5); (**b,c**) annual average hydropower availability for storage (**b**) and diversion (**c**) sites under SSP1-2.6 from HydroSMADE, averaged across the 10 GCMs used in v1.0; and (**d**) annual average streamflow under SSP1-2.6 from HydroSMADE, also averaged across the same 10 GCMs. Note that the historical and future periods are aligned with the availability of the Gernaat *et al.* dataset, which is available for 158 out of the 235 global basins (panel a).

No globally consistent plant-level monthly reference exists for the sites represented in HydroSMADE. We therefore compare HydroSMADE with RectifHydPlus_CTRL, the hydrological-control series within the RectifHydPlus[16], a monthly hydropower reanalysis (1980–2019) for existing sites in the contiguous United States. RectifHydPlus_CTRL reduces the influence of historical capacity changes and operational disruptions by modeling annual capacity factors from water availability, applying contemporary nameplate capacity, and distributing annual generation among months using plant- and year-specific water proxies. This reanalysis is thus more comparable to HydroSMADE's hydrologically conditioned potential than a record of observed monthly generation.

Matching names and locations between the two products yields 123 common facilities (Supplementary Table S3). For each product and facility-year, we express each month as a percentage of annual generation and average these shares by calendar month over 1980–2019; HydroSMADE results are also averaged across 30 scenarios. This removes annual magnitude and simple capacity scaling and focuses the comparison on the typical within-year distribution. We evaluate the resulting profiles using Root Mean Square Error (RMSE), which measures month-specific differences in percentage points, and Pearson correlation coefficient (r), which measures seasonal covariation rather than equality. Supplementary Texts S3 and S4 provide metric equations for r and RMSE, along with the details of the spatial and identifier matching procedure used to align facilities across datasets.

Median RMSE is approximately 4 percentage points (interquartile range: 2.5–6; full range: 1–15), while median r is approximately 0.42 (interquartile range: 0.03–0.59; full range: −0.66–0.96; Fig. 8). Taken together, these metrics indicate that HydroSMADE provides a broadly reasonable representation of the typical within-year distribution at many facilities, but with substantial site-to-site variation, particularly in seasonal timing. The results therefore support its intended broad-scale application but not unqualified use of its monthly profiles for individual-facility analysis.

To illustrate what these metric combinations represent in the underlying monthly profiles, we divide the facilities into three descriptive categories. Of the 123 facilities, 62 have positive r and RMSE below 5 percentage points, indicating a positive seasonal association with comparatively small month-specific differences. Another 32 have positive r and RMSE of at least 5 percentage points, indicating that seasonal highs and lows generally occur at similar times but that the shares assigned to individual months differ more substantially. The remaining 29 have negative r, indicating different relative timing of high- and low-share months. Thus, 94 of 123 facilities show positive seasonal covariation, although its strength and the magnitude of the monthly-share differences vary. The 5-percentage-point cutoff organizes the profile comparisons and is not a threshold for acceptable performance.

Figure 9 shows representative profiles from all three categories. The first row illustrates agreement in seasonal timing and monthly-share magnitude; the second shows that similar seasonal ordering can coexist with larger

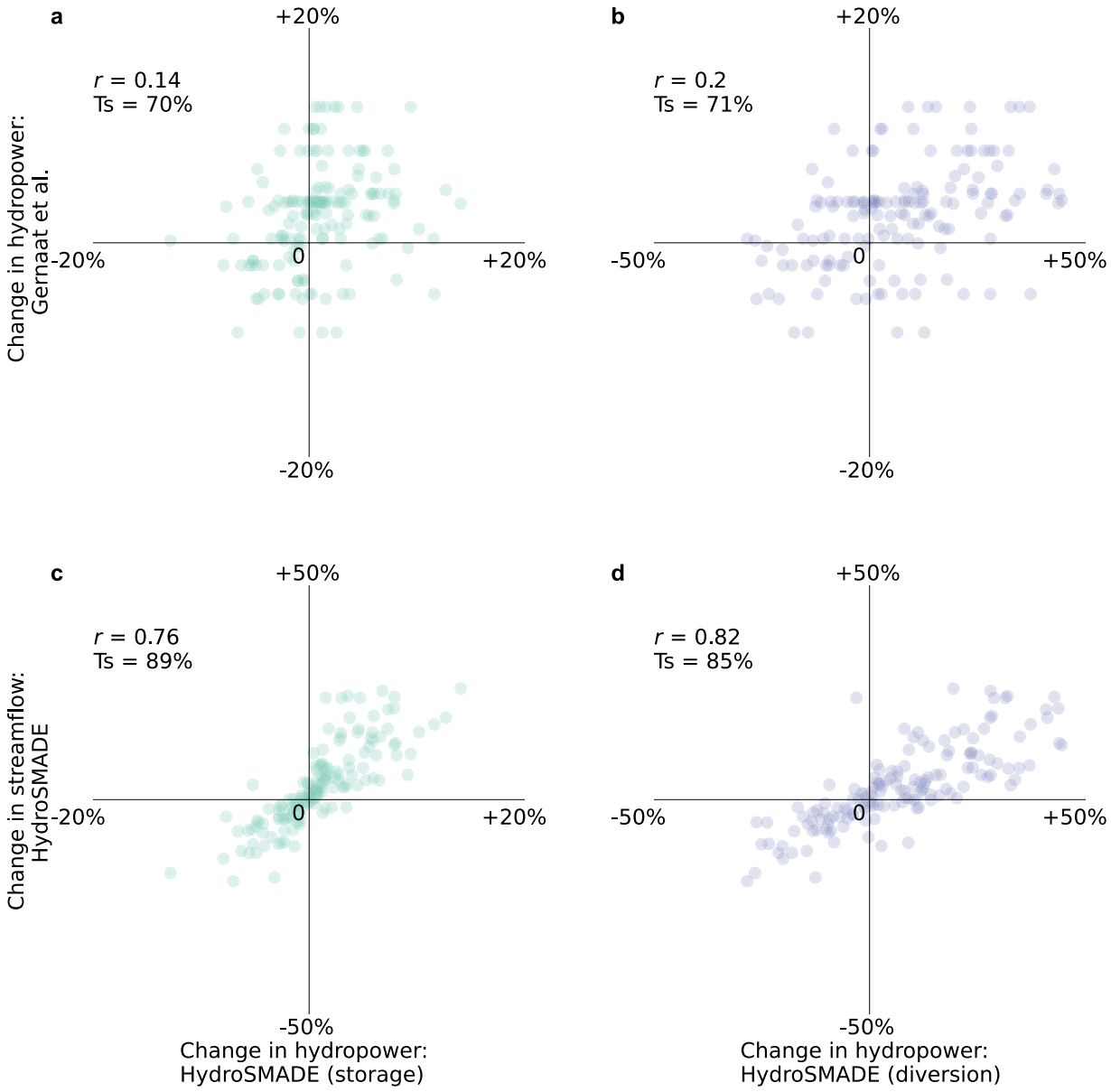
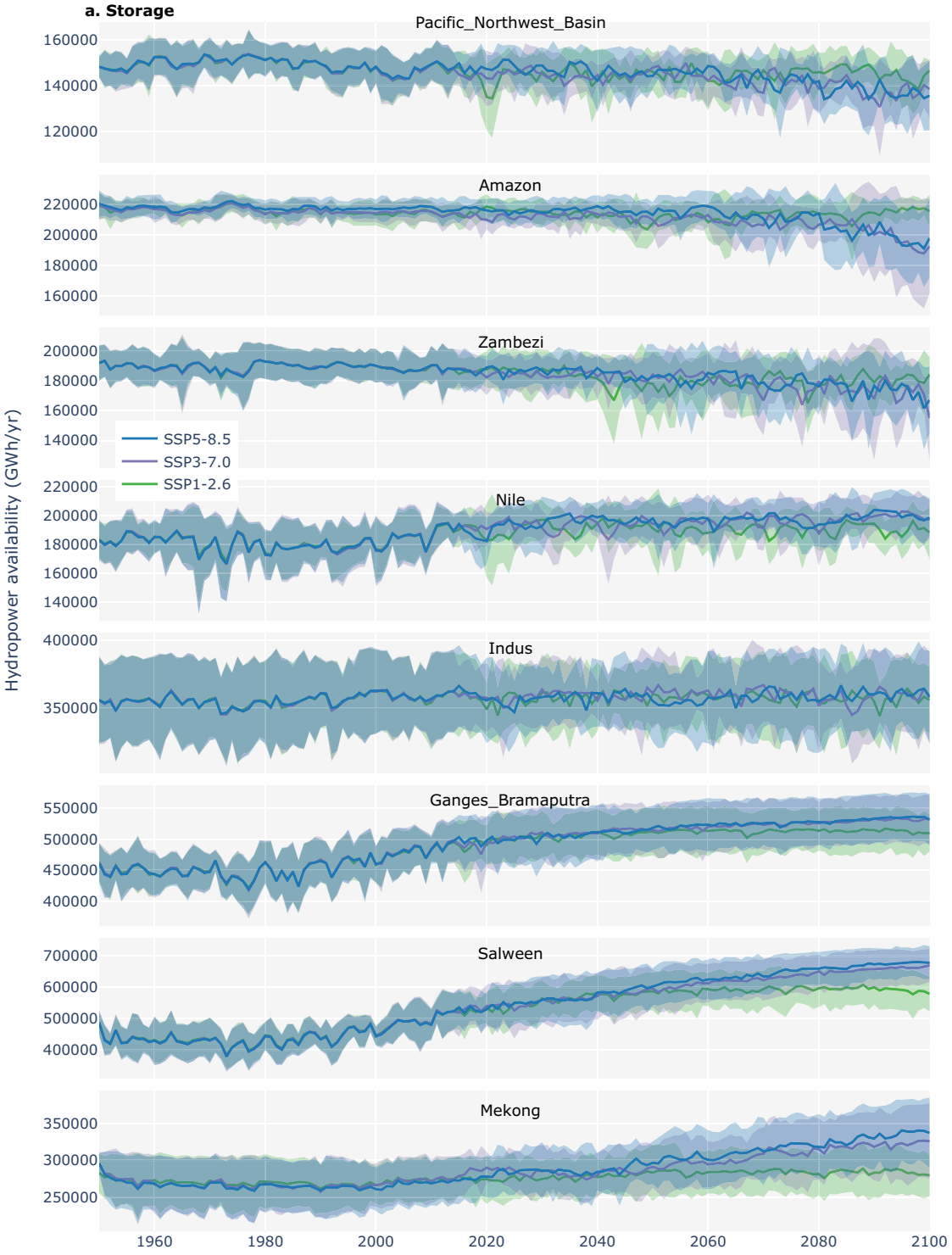


Fig. 5 Comparison of projected percentage changes in annual mean hydropower availability in HydroSMADE versus that in Gernaat *et al.* and HydroSMADE's annual mean streamflow. For 158 river basins with availability of Gernaat *et al.* data, future changes in hydropower availability in HydroSMADE versus: (**a,b**) Gernaat *et al.* for storage (**a**) and diversion (**b**) sites, respectively; (**c,d**) HydroSMADE streamflow for storage (**c**) and diversion (**d**) sites. All changes are expressed as the relative difference between a future period (2070–2100) and a historical baseline (1970–2000). Gernaat *et al.* statistics are averaged across four GCMs (GFDL-ESM2M, HadGEM2-ES, IPSL-CM5A-LR, and MIROC5) under RCP2.6, whereas HydroSMADE statistics are averaged across the 10 GCMs under SSP1-2.6 used in v1.0. 'Ts' represents the share of basins for which HydroSMADE shows similar future trajectories as Gernaat *et al.* (**a,b**) and streamflow (**c,d**), whereas '*r*' represents the Pearson correlation coefficient between the pair of statistics.

month-specific differences; and the third shows profiles placing their relative seasonal highs and lows in different months. Expanded comparisons are provided in Supplementary Figs. S10–S12.

The remaining differences should be interpreted in light of the distinct inputs and purposes of the two products. RectifHydPlus_CTRL uses plant- and year-specific water proxies, whereas HydroSMADE simulates monthly reservoir operations under an annual energy-maximization objective and assigns a common per-unit-capacity monthly profile to facilities of the same modeled type within a grid cell. Differences may therefore arise from hydrologic forcing, plant and reservoir attributes, grid-to-site disaggregation, and the representation of monthly operations.

Because these factors cannot be controlled independently and RectifHydPlus_CTRL is itself a reanalysis, agreement does not constitute definitive validation of HydroSMADE, and disagreement cannot be attributed


a. Storage
Pacific_Northwest_Basin
Amazon
Zambezi
Nile
Indus
Ganges_Bramaputra
Salween
Mekong
Hydropower availability (GWh/yr)
SSP5-8.5
SSP3-7.0
SSP1-2.6
160000
140000
120000
220000
200000
180000
160000
200000
180000
160000
140000
220000
200000
180000
160000
140000
400000
350000
550000
500000
450000
400000
700000
600000
500000
400000
350000
300000
250000
1960
1980
2000
2020
2040
2060
2080
2100

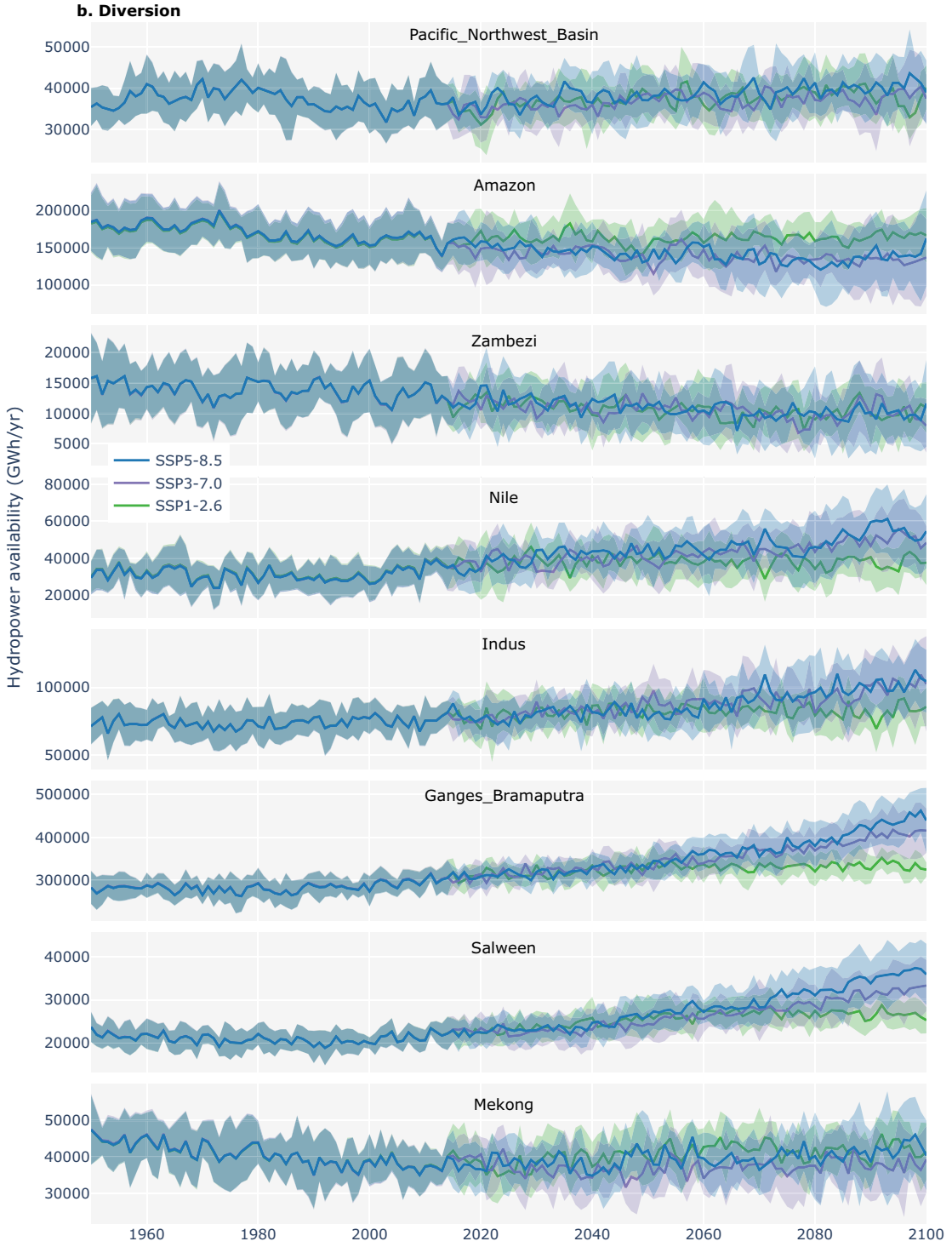


Fig. 6 Basin-scale annual hydropower availability in (**a**) storage and (**b**) diversion-type hydropower sites at eight representative basins over 1950–2100 under three alternative futures. For each alternative future, the solid line and error band represent the mean and standard deviation of hydropower availability for 10 GCMs.

uniquely to an error in either product. This comparison should therefore be interpreted as one complementary, product-level benchmark of HydroSMADE's typical within-year distribution. It does not validate historical dispatch, absolute annual generation, capacity factor, maximum monthly availability, or individual reservoir operations.

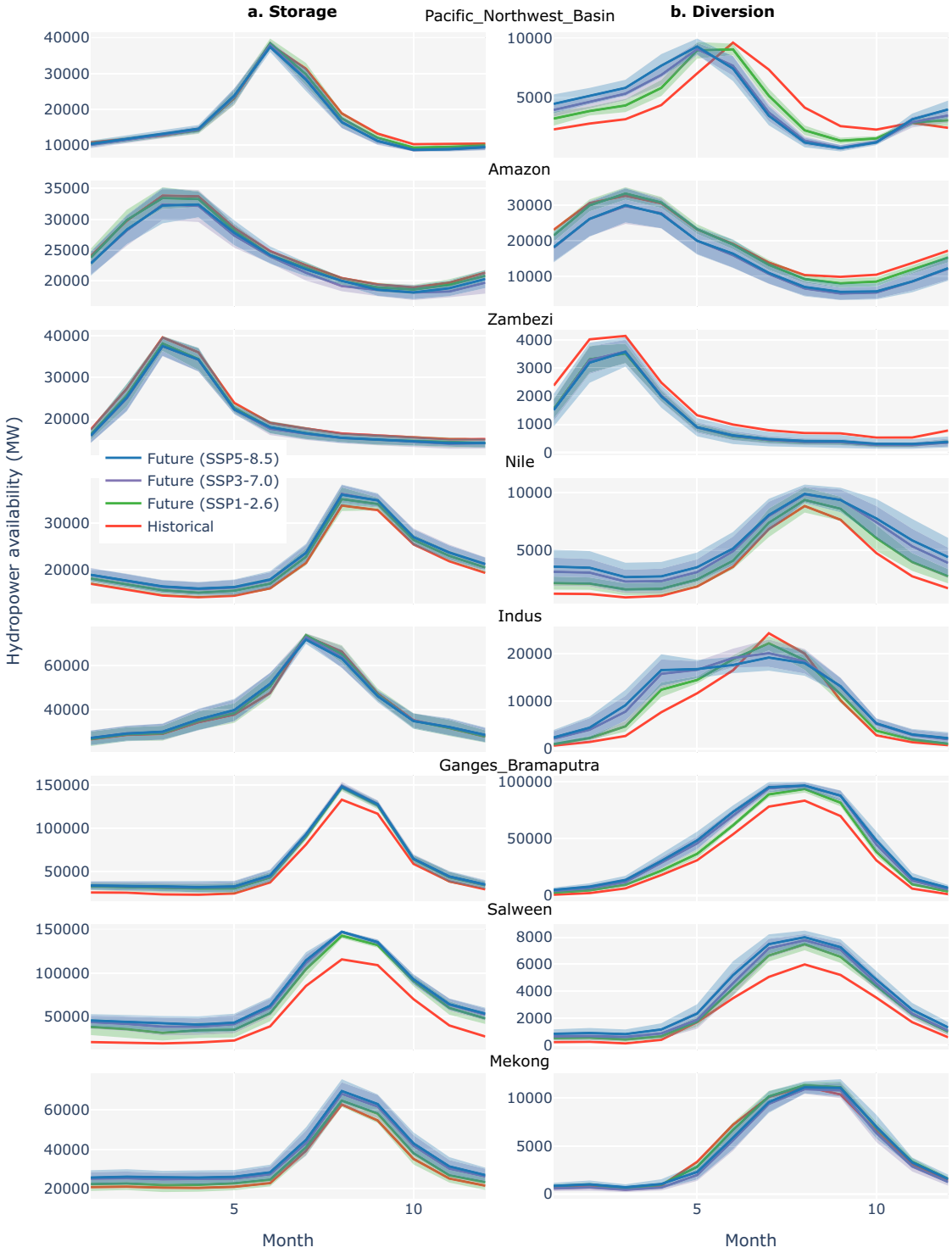


Fig. 7 Basin-scale monthly mean hydropower availability in (**a**) storage-type and (**b**) diversion dams at eight representative basins for a historical (1970–2000) and a future (2070–2100) period under three alternative futures. For each alternative future, the solid line and error bar represent the mean and standard deviation of the hydropower availability for 10 GCMs.

Usage Notes

Potential applications. HydroSMADE v1.0 provides site-level estimates of technically achievable monthly hydropower availability under modeled, energy-maximizing operations across alternative future hydrologic scenarios. Here, "availability" refers to the hydropower that could technically be produced in each month under the modeled conditions. It does not represent actual historical or future dispatch, firm energy, dispatchable capacity, installed capacity, or undeveloped potential capacity.

Basin	Change (%) in hydropower in Gernaat et al. (RCP2.6 mean)	Change (%) in hydropower in HydroSMADE				Change (%) in streamflow (HydroSMADE)	
		SSP1-2.6 mean (storage)	SSP1-2.6 mean (diversion)	Ensemble mean (storage)	Ensemble mean (diversion)	SSP1-2.6 mean	Ensemble mean
Amazon	−0.3	−0.8	−0.9	−4	−11	−5	−20
Ganges Bramaputra	4	14	19	16	35	26	30
Indus	5	0.6	12	1	27	2	4
Mekong	3	6	5	19	2	5	2
Nile	6	6	25	10	58	10	15
Pacific Northwest Basin	4	−3	1	−6	2	−5	−7
Salween	3	36	32	47	49	24	12
Zambezi	−5	−5	−26	−7	−25	−5	−10

Table 3. For the eight representative basins, comparison of percentage changes in projected annual mean hydropower availability in HydroSMADE against the hydropower availability in Gernaat *et al.* and HydroSMADE's annual mean streamflow. The statistics show the relative changes in a future period (2070–2100) compared to a historical baseline (1970–2000). Gernaat *et al.* statistics are averaged across four GCMs (GFDL-ESM2M, HadGEM2-ES, IPSL-CM5A-LR, and MIROC5) under RCP2.6. HydroSMADE statistics in the third, fourth, and seventh columns are averaged across the 10 GCMs under SSP1-2.6, where those in the fifth, sixth, and eighth columns are averaged across the 30-scenario ensembles.

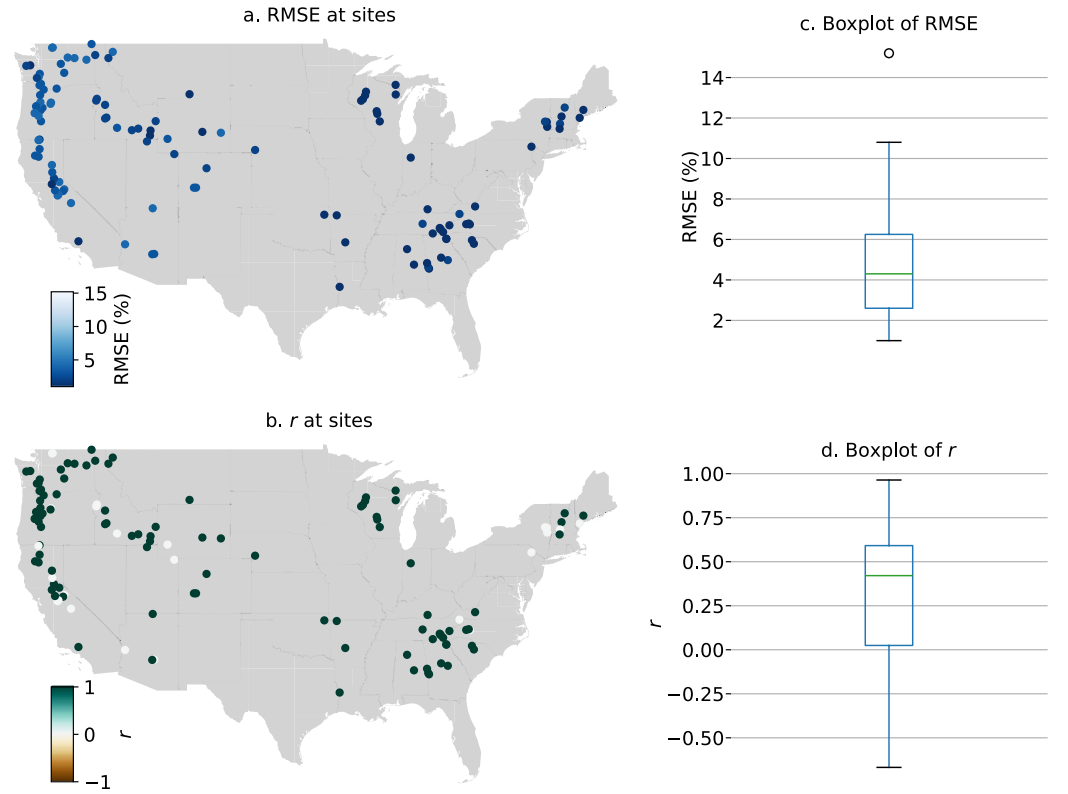


Fig. 8 Site-level comparison of how HydroSMADE and RectifHydPlus_CTRL distribute annual hydropower generation across the 12 calendar months at 123 matched existing facilities in the contiguous United States. For each dataset, monthly shares were calculated separately for each facility and year and then averaged by calendar month over 1980–2019; the HydroSMADE shares were also averaged across the 30 scenarios. Panels (**a**) and (**b**) map the resulting facility-level RMSE and Pearson correlation coefficient, *r*, respectively, and panels (**c**) and (**d**) show their distributions across the 123 facilities. RMSE is reported in percentage points of annual generation, with lower values indicating smaller differences between the two monthly distributions. *r* is dimensionless, with positive values indicating that relatively high- and low-share months tend to occur at similar times in the two datasets and negative values indicating opposing seasonal patterns.

The technical validation supports the use of HydroSMADE to characterize broad spatial patterns, scenario uncertainty, and the typical within-year distribution of hydropower availability. The dataset is therefore intended primarily for aggregated subnational, national, continental, and global applications. Potential uses include

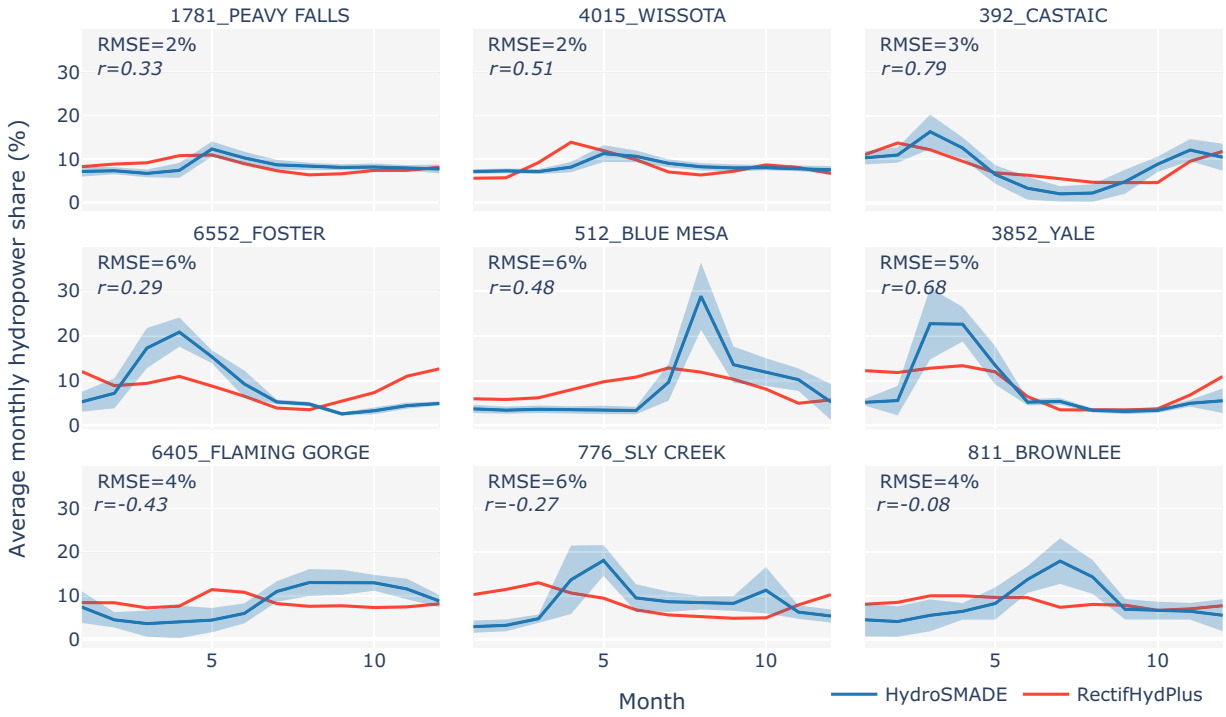


Fig. 9 Representative average monthly hydropower-share profiles for the three descriptive categories. Rows from top to bottom correspond to the three categories: positive *r* with RMSE below 5 percentage points, positive *r* with RMSE of at least 5 percentage points, and negative *r*. Columns from left to right show sites nearest the 25th, 50th, and 75th percentiles of *r* within each category. HydroSMADE shading denotes ± 2 standard deviations across 30 scenarios. Plot titles provide RectifHydPlus facility identifiers (RHPID); corresponding HydroSMADE identifiers, locations, and HydroSMADE capacities are provided in Supplementary Table S3. Results for additional sites are shown in Supplementary Figs. S10–S12.

evaluating the future role and seasonality of hydropower in power-system planning and multisector models, comparing hydropower availability across alternative hydroclimatic and hydrologic scenarios, and screening prospective hydropower sites or regions for further investigation. Prospective-site estimates should be treated as screening-level technical information rather than evidence of project-specific economic viability, environmental acceptability, permitting feasibility, or operational performance.

Limitations. HydroSMADE should not be interpreted as a reconstruction or prediction of individual-facility operations. Actual hydropower generation may differ because of plant-specific operating objectives, outages, aging and maintenance, electricity-system conditions, environmental regulations, competing water uses, and basin-level water-management decisions. Accordingly, facility-level monthly profiles require additional local information and evaluation before use in operational planning, project design, licensing, resource-adequacy assessment, or other applications requiring plant-specific accuracy.

The 0.5° Xanthos representation aggregates facilities of the same modeled type within a grid cell into a representative grid-cell facility rather than simulating each facility separately within an explicit river network. Following simulation, the grid-to-site disaggregation procedure assigns a common per-unit-capacity monthly profile to facilities of the same modeled type within a grid cell. This procedure preserves the simulated grid-cell total but does not fully represent site-specific differences in hydraulic head, storage, efficiency, turbine-flow limits, operating rules, or cascade position. The monthly comparison with RectifHydPlus_CTRL supports the broad-scale representation of typical within-year hydropower availability, but agreement does not hold uniformly at individual facilities, where the products can differ in the magnitude of monthly shares and the timing of seasonal highs and lows. Because the two products differ in hydrologic forcing, plant and reservoir attributes, grid-to-site disaggregation, and representations of monthly operations, their differences cannot be attributed uniquely to any one component of HydroSMADE. Future work should conduct a systematic sensitivity analysis of the principal HydroSMADE assumptions, including reservoir-attribute aggregation, coarse-grid versus explicit vector-network representation of individual and cascaded reservoirs, operating-rule formulation, and grid-to-site disaggregation, to quantify their respective contributions to uncertainty in site-level monthly availability. Finally, HydroSMADE includes conventional storage and diversion hydropower sites but does not represent pumped-storage hydropower, which also has substantial untapped potential worldwide[75,76].

Data availability

The HydroSMADE v1.0 dataset is publicly available on MSD LIVE at https://doi.org/10.57931/3015845.

Code availability

The version of Xanthos (v2.4.1) used in producing HydroSMADE v1.0 is available at https://github.com/JGCRI/xanthos. We also provide example Jupyter Notebooks to explore HydroSMADE at https://github.com/kamal0013/HydroSMADE.

Received: 17 February 2026; Accepted: 8 October 2026;
Published online: 31 October 2026

References

1. Liu, Z. & He, X. Balancing-oriented hydropower operation makes the clean energy transition more affordable and simultaneously boosts water security. *Nat. Water* **1**, 778–789 (2023).
2. Bracken, C. *et al*. Seasonal compound renewable energy droughts in the United States. *Environ. Res. Energy* **2**, 025005 (2025).
3. Bracken, C., Son, Y., Broman, D. & Voisin, N. GODEEEP-hydro: Historical and projected power system ready hydropower data for the United States. *Sci. Data* **12**, 875 (2025).
4. Turner, S. W. D. & Voisin, N. Simulation of hydropower at subcontinental to global scales: a state-of-the-art review. *Environ. Res. Lett.* **17**, 023002 (2022).
5. Arnold, W., Giuliani, M. & Castelletti, A. Floating photovoltaics may reduce the risk of hydro-dominated energy development in Africa. *Nat. Energy* **9**, 602–611 (2024).
6. Magee, T. M. *et al*. Evaluating power grid model hydropower feasibility with a river operations model. *Environ. Res. Lett.* **17**, 084035 (2022).
7. Ploussard, Q., Veselka, T. D., Oikonomou, K. & Voisin, N. Hydro-economics tradeoff surfaces to guide unit commitment in production cost models. *Appl. Energy* **324**, 119728 (2022).
8. Carlino, A., Schmitt, R., Clark, A. & Castelletti, A. Rethinking energy planning to mitigate the impacts of African hydropower. *Nat. Sustain.* **7**, 879–890 (2024).
9. Chowdhury, A. F. M. K., Wessel, J., Wild, T., Lamontagne, J. & Kanyako, F. Exploring sustainable electricity system development pathways in South America's MERCOSUR sub-region. *Energy Strategy Rev.* **49**, 101150 (2023).
10. Shah, J., Hu, J., Edelenbosch, O. Y. & van Vliet, M. T. H. Global dataset combining open-source hydropower plant and reservoir data. *Sci. Data* **12**, 646 (2025).
11. Lehner, B. *et al*. The Global Dam Watch database of river barrier and reservoir information for large-scale applications. *Sci. Data* **11**, 1069 (2024).
12. Zhang, A. T. & Gu, V. X. Global Dam Tracker: A database of more than 35,000 dams with location, catchment, and attribute information. *Sci. Data* **10**, 111 (2023).
13. Steyaert, J. C., Condon, L. E., Sean, W. D. T. & Voisin, N. ResOpsUS, a dataset of historical reservoir operations in the contiguous United States. *Sci. Data* **9**, 34 (2022).
14. Wan, W., Zhao, J., Popat, E., Herbert, C. & Döll, P. Analyzing the Impact of Streamflow Drought on Hydroelectricity Production: A Global-Scale Study. *Water Resour. Res.* **57**, e2020WR028087 (2021).
15. Lehner, B. *et al*. High-resolution mapping of the world's reservoirs and dams for sustainable river-flow management. *Front. Ecol. Environ.* **9**, 494–502 (2011).
16. Turner, S. W. D., Singh, D., Hansen, C. & Kao, S.-C. Forty-year hydropower generation reanalysis for Conterminous United States. *Sci. Data* **12**, 1040 (2025).
17. Turner, S. W. D., Voisin, N. & Nelson, K. Revised monthly energy generation estimates for 1,500 hydroelectric power plants in the United States. *Sci. Data* **9**, 675 (2022).
18. Son, Y., Bracken, C., Broman, D. & Voisin, N. Monthly hydropower generation data for Western Canada to support Western-US interconnect power system studies. *Sci. Data* **12**, 874 (2025).
19. Zhao, Y. *et al*. Future precipitation, hydrology and hydropower generation in the Yalong River Basin: Projections and analysis. *J. Hydrol.* **602**, 126738 (2021).
20. Pöyry & ECREEE. *GIS Hydropower Resource Mapping – Country Report for Mali* (2017).
21. van Dijk, M., Anja, B. & Chantel, N. *Development of the South African Hydropower Atlas* (2021).
22. Sammartano, V., Liuzzo, L. & Freni, G. Identification of Potential Locations for Run-of-River Hydropower Plants Using a GIS-Based Procedure. *Energies* **12**, 3446 (2019).
23. Mudenda, F., van Dijk, M. & Bekker, A. Development of evaluation framework for the selection of run-of-river hydropower potential sites to be included in the Zambian Hydropower Atlas. *J. Water Clim. Change* **13**, 4000–4018 (2022).
24. The World Bank. *Small Hydro Resource Mapping in Tanzania: Phase 1 Report* (2015).
25. The World Bank. *Small Hydro Resource Mapping in Tanzania: Final Report* (2018).
26. The World Bank. *Small Hydro Resource Mapping in Madagascar: Executive Summary* (2017).
27. The World Bank. *Hydro Atlas of Madagascar* (2017).
28. Chowdhury, A. F. M. K. *et al*. Hydropower expansion in eco-sensitive river basins under global energy-economic change. *Nat. Sustain.* 1–10 https://doi.org/10.1038/s41893-023-01260-z (2024).
29. Zhang, Y. *et al*. Long-term basin-scale hydropower expansion under alternative scenarios in a global multisector model. *Environ. Res. Lett.* **17**, 114029 (2022).
30. Abeshu, G. W. *et al*. Enhancing the representation of water management in global hydrological models. *Geosci. Model Dev.* **16**, 5449–5472 (2023).
31. Li, X. *et al*. Xanthos – A Global Hydrologic Model. *J. Open Res. Softw.* **5** (2017).
32. Vernon, C. R. *et al*. A Global Hydrologic Framework to Accelerate Scientific Discovery. *J. Open Res. Softw.* **7**, 1 (2019).
33. Liu, Y., Hejazi, M., Li, H., Zhang, X. & Leng, G. A hydrological emulator for global applications – HE v1.0.0. *Geosci. Model Dev.* **11**, 1077–1092 (2018).
34. Xu, R. *et al*. A global-scale framework for hydropower development incorporating strict environmental constraints. *Nat. Water* **1**, 113–122 (2023).
35. Turner, S. W. D., Ng, J. Y. & Galelli, S. Examining global electricity supply vulnerability to climate change using a high-fidelity hydropower dam model. *Sci. Total Environ.* **590–591**, 663–675 (2017).
36. McManamay, R. A., Oigbokie, C. O., Kao, S.-C. & Bevelhimer, M. S. Classification of US Hydropower Dams by their Modes of Operation. *River Res. Appl.* **32**, 1450–1468 (2016).
37. Fantin-Cruz, I., Pedrollo, O., Girard, P., Zeilhofer, P. & Hamilton, S. K. Effects of a diversion hydropower facility on the hydrological regime of the Correntes River, a tributary to the Pantanal floodplain, Brazil. *J. Hydrol.* **531**, 810–820 (2015).
38. Global Energy Monitor. Global Hydropower Tracker (2025).

39. Lange, S. Trend-preserving bias adjustment and statistical downscaling with ISIMIP3BASD (v1.0). *Geosci. Model Dev.* **12**, 3055–3070 (2019).
40. Zhao, M., Wild, T. & Vernon, C. Xanthos Output Dataset Under ISIMIP3b Selected CMIP6 Scenarios: 1850–2100. MultiSector Dynamics - Living, Intuitive, Value-adding, Environment, https://doi.org/10.57931/2217355 (2023).
41. Calvin, K. *et al*. GCAM v5.1: representing the linkages between energy, water, land, climate, and economic systems. *Geosci. Model Dev.* **12**, 677–698 (2019).
42. Khan, Z. *et al*. Global monthly sectoral water use for 2010–2100 at 0.5° resolution across alternative futures. *Sci. Data* **10**, 201 (2023).
43. Niazi, H. *et al*. Global peak water limit of future groundwater withdrawals. *Nat. Sustain.* **7**, 413–422 (2024).
44. Graham, N. T. *et al*. Agricultural market integration preserves future global water resources. *One Earth* **6**, 1235–1245 (2023).
45. Zhao, M. *et al*. GCAM–GLORY v1.0: representing global reservoir water storage in a multi-sector human–Earth system model. *Geosci. Model Dev.* **17**, 5587–5617 (2024).
46. Graham, N. T. *et al*. Humans drive future water scarcity changes across all Shared Socioeconomic Pathways. *Environ. Res. Lett.* **15**, 014007 (2020).
47. Kanyako, F., Lamontagne, J., Baker, E., Turner, S. & Wild, T. Seasonality and trade in hydro-heavy electricity markets: A case study with the West Africa Power Pool (WAPP). *Appl. Energy* **329**, 120214 (2023).
48. Jin, X., Chowdhury, A. F. M. K., Liu, B., Cheng, C. & Galelli, S. China Southern Power Grid's decarbonization likely to impact cropland and transboundary rivers. *Commun. Earth Environ.* **5**, 1–11 (2024).
49. Wild, T. B. *et al*. Integrated energy-water-land nexus planning in the Colorado River Basin (Argentina). *Reg. Environ. Change* **21**, 62 (2021).
50. Khan, Z. *et al*. Integrated energy-water-land nexus planning to guide national policy: an example from Uruguay. *Environ. Res. Lett.* **15**, 094014 (2020).
51. Wild, T. B. *et al*. The Implications of Global Change for the Co-Evolution of Argentina's Integrated Energy-Water-Land Systems. *Earths Future* **9**, e2020EF001970 (2021).
52. Thomas, H. A. *Improved Methods for National Water Assessment, Water Resources Contract: WR15249270*. https://pubs.usgs.gov/publication/70046351, https://doi.org/10.3133/70046351 (1981).
53. Branstetter, M. L. & Erickson III, D. J. Continental runoff dynamics in the Community Climate System Model 2 (CCSM2) control simulation. *J. Geophys. Res. Atmos.* **108** (2003).
54. Zhou, Y. *et al*. A comprehensive view of global potential for hydro-generated electricity. *Energy Environ. Sci.* **8**, 2622–2633 (2015).
55. Loucks, D. P. & van Beek, E. An Introduction to Optimization Models and Methods. in *Water Resource Systems Planning and Management: An Introduction to Methods, Models, and Applications* (eds Loucks, D. P. & van Beek, E.) 93–177 https://doi.org/10.1007/978-3-319-44234-1_4 (Springer International Publishing, Cham, 2017).
56. Hanasaki, N. *et al*. An integrated model for the assessment of global water resources – Part 1: Model description and input meteorological forcing. *Hydrol. Earth Syst. Sci.* **12**, 1007–1025 (2008).
57. Müller Schmied, H. *et al*. The global water resources and use model WaterGAP v2.2d: model description and evaluation. *Geosci. Model Dev.* **14**, 1037–1079 (2021).
58. Chowdhury, A. K. *et al*. Repository of HydroSMADE: Hydropower Site-level Monthly Availability Data Ensemble for 1950–2100 at Existing and Potential Global Sites. Pacific Northwest National Lab (United States), https://doi.org/10.57931/3015845.
59. Müller Schmied, H. *et al*. The global water resources and use model WaterGAP v2.2e: description and evaluation of modifications and new features. *Geosci. Model Dev.* **17**, 8817–8852 (2024).
60. Gupta, H. V., Kling, H., Yilmaz, K. K. & Martinez, G. F. Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *J. Hydrol.* **377**, 80–91 (2009).
61. Knoben, W. J. M., Freer, J. E. & Woods, R. A. Technical note: Inherent benchmark or not? Comparing Nash–Sutcliffe and Kling–Gupta efficiency scores. *Hydrol. Earth Syst. Sci.* **23**, 4323–4331 (2019).
62. Lin, P. *et al*. Global Reconstruction of Naturalized River Flows at 2.94 Million Reaches. *Water Resour. Res.* **55**, 6499–6516 (2019).
63. Kaunda, C. S., Kimambo, C. Z. & Nielsen, T. K. Hydropower in the Context of Sustainable Energy Supply: A Review of Technologies and Challenges. *Int. Sch. Res. Not.* **2012**, 730631 (2012).
64. Gernaat, D. E. H. J. *et al*. Climate change impacts on renewable energy supply. *Nat. Clim. Change* **11**, 119–125 (2021).
65. Hamlet, A. *et al*. Effects of climate change on water resources in the Pacific Northwest: Impacts and policy implications. http://hdl.handle.net/1773/34506 (2001).
66. Voisin, N., Gore, S. & Kao, S.-C. *Balancing Hydropower Generation, Flood Control and Drought Management Services Under a Changing Climate* (2022).
67. Broman, D., Voisin, N., Kao, S.-C., Fernandez, A. & Ghimire, G. R. Multi-scale impacts of climate change on hydropower for long-term water-energy planning in the contiguous United States. *Environ. Res. Lett*. https://doi.org/10.1088/1748-9326/ad6ceb (2024).
68. Almeida, R. M. *et al*. Climate change may impair electricity generation and economic viability of future Amazon hydropower. *Glob. Environ. Change* **71**, 102383 (2021).
69. Arias, M., Farinosi, F. & Hughes, D. Future hydropower operations in the Zambezi River Basin: Climate impacts and adaptation capacity. *RIVER Res. Appl*. https://doi.org/10.1002/rra.3958 (2022).
70. Wasti, A. *et al*. Climate change and the hydropower sector: A global review. *WIREs Clim. Change* **13**, e757 (2022).
71. Ly, S., Sayama, T. & Try, S. Integrated impact assessment of climate change and hydropower operation on streamflow and inundation in the lower Mekong Basin. *Prog. Earth Planet. Sci.* **10**, 55 (2023).
72. Su, F. *et al*. Hydrological response to future climate changes for the major upstream river basins in the Tibetan Plateau. *Glob. Planet. Change* **136**, 82–95 (2016).
73. Wijngaard, R. R. *et al*. Future changes in hydro-climatic extremes in the Upper Indus, Ganges, and Brahmaputra River basins. *PLOS ONE* **12**, e0190224 (2017).
74. Stedinger, J. R. & Taylor, M. R. Synthetic streamflow generation: 1. Model verification and validation. *Water Resour. Res.* **18**, 909–918 (1982).
75. Stocks, M., Stocks, R., Lu, B., Cheng, C. & Blakers, A. Global Atlas of Closed-Loop Pumped Hydro Energy Storage. *Joule* **5**, 270–284 (2021).
76. Weber, T., Stocks, R., Blakers, A., Nadolny, A. & Cheng, C. A global atlas of pumped hydro systems that repurpose existing mining sites. *Renew. Energy* **224**, 120113 (2024).

Acknowledgements

This research is supported by the US Department of Energy, Office of Science, as part of research in Multi-Sector Dynamics, Earth and Environmental System Modeling Program (contract number DE-AC05-76RL01830). The authors acknowledge Dr. Rongrong Xu for sharing the attribute data of the potential hydropower sites. The views and opinions expressed in this paper are those of the authors alone.

Author contributions

A.F.M.K.C. and T.W. conceptualized the study; G.W.A. and M.Z. set up the Xanthos model and curated its input data; A.F.M.K.C. disaggregated the Xanthos output from grid-scale to site-scale, prepared the final HydroSMADE product, and conducted the analysis; A.F.M.K.C., G.W.A., M.Z., H.Y.L., and T.W. wrote the first draft; all the co-authors contributed to the writing and discussion.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41597-026-08515-2.

Correspondence and requests for materials should be addressed to A.F.M.K.C.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

© Battelle Memorial Institute and A. F. M Kamal Chowdhury, Gi Joo Kim, Jonathan Lamontagne, Hong-Yi Li 2026

This Article in Press is shared early to give you faster access to new research. It is citable and carries a permanent DOI. The final edited version will replace it automatically.